Q1

$a = 4$, $b = 3/8$, $c = -2$

Q2

$x = [4 \pm \sqrt{(16 + 12(2+\sqrt{5})(2-\sqrt{5}))}] / [2(2+\sqrt{5})]$. Since $(2+\sqrt{5})(2-\sqrt{5}) = 4-5 = -1$,

this simplifies to $x = (4 \pm \sqrt{(16-12)}) / (2(2+\sqrt{5})) = (4 \pm 2) / (2(2+\sqrt{5}))$

Q3a

leading coefficient = +3, one factor = $(x+2)$, remaining factors = $(x-1)(x-4)$

Q3b

Case 1: $5x-2 \geq 4x+1$ gives $x \geq 3$ (critical value 3). Case 2: $5x-2 \leq -4x-1$ gives $9x \leq 1$ so

$x \leq 1/9$ (critical value 1/9). Combined solution: $1/9 \leq x \leq 3$

Q4a

$r\theta = 12$ and $\frac{1}{2}r^2\theta = 57.6$. Dividing the second by the first: $\frac{1}{2}r = 57.6/12 = 4.8$, so r

$= 9.6$. Then $\theta = 12/9.6 = 1.25$

Q4b

$AC = r\tan\theta = 9.6 \times \tan(1.25) = 28.89$. Shaded area = $(\frac{1}{2} \times 28.89 \times 9.6) - 57.6 =$

$138.67 - 57.6 = 81.1$